



Homer Gage Balcom and the Empire State Building

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The ultimate masterpiece of the many remarkable skyline buildings designed by New York structural engineer Homer "H. G." Balcom (Fig. 1) during the early to middle part of the 20th century was the incomparable Empire State Building (Fig. 2). Completed in record-setting time, just 13 months from start to finish for construction, Balcom's work of genius was the tallest skyscraper ever built and the world record holder for more than four decades, from 1931 to 1972.

Immediately after the historic steel-frame, masonry-walled building debuted on May 1, 1931, it was hailed as the Eighth Wonder of the World, and it became (and still is) one of the most recognizable skyscrapers in the world, an iconic landmark of New York City and America. The early 20th-century Empire State Building



Figure 1. Homer "H. G." Balcom (Photo courtesy of Lois Moon Welch)

revealed to the rest of the world that within the structural engineering community of the United States, there was a high level of daring, ingenuity, and imagination. Without question, American engineers had developed the know-how and expertise to lead the world in the design and construction of increasingly higher record-setting buildings.

Recognized internationally as a pioneering expert for his increasingly bold use of steel and wrought iron in complex structures, Balcom advanced numerous innovative design concepts that made very tall buildings stable and safe when exposed to extremely large lateral forces. In addition to being a world authority on the effects of wind stresses on high-rises, Balcom was also expert at designing unique



Figure 2. Empire State Building, New York (Photo courtesy of Evelyn Weingardt)

foundations that eliminated vibrations in all types of building structures. Many of the creative methods he advanced regarding design standards, engineering techniques, and construction procedures remain applicable today.

In his seminal paper on the construction of the Empire State Building, "New York's Tallest Skyscraper," Balcom (1931) wrote,

The building exemplifies the fact that good construction and good architecture can go hand in hand if sympathetic cooperation exists between the architect and the engineer. Such cooperation is, in fact, of the utmost importance in producing satisfactory results; and, as architects are giving this subject more and more attention, it is of first importance that structural engineers render construction assistance in the field.

He ended his paper with these words: "It is, therefore, greatly to be hoped that this building may be noted, not only for its height and its majestic beauty, but also as a means of promoting engineering knowledge" (Balcom 1931).

ASCE's introduction to Balcom's paper read,

Since the dawn of civilization it has ever been the dream of man to build lofty structures. In ancient times, this was exemplified in the building of the Hanging Gardens of Babylon, the Colossus of Rhodes, and the pyramids of Egypt. In medieval times, and until the latter

part of the last century, the building of great cathedrals of the old world marked the highest achievement in building construction. All of these structures were limited by the materials then available for use. Within the last thirty years, however, the development of structural steel has opened the field of almost unlimited possibilities and has led to the construction of buildings that surpass in height anything before attained. The most recent of these great structures is the Empire State Building. (Balcom 1931)

Over its life, the Empire State Building has been the focus of numerous headline-grabbing events and the setting for many popular Hollywood movies. One of the first was the 1933 *King Kong*, in which a lovesick supergiant gorilla and his hysterical blond female captive are pursued to the top of the building, where King Kong meets his demise, shot down by scores of fighter planes.

The building's most structurally damaging incident occurred on a foggy Saturday morning in July 1945. A pilot headed for Newark airport lost his way and crashed his B-25 bomber into the structure's 79th floor, killing himself and 13 others. Although the building suffered considerable fire damage and a gaping 20-foot gash in its side, it was quickly repaired and open for business on the following Monday.

Balcom's bulging portfolio of outstanding projects included numerous landmark buildings and bridges built around the United States and in Europe. His list of imposing structures commenced with the design of the mammoth Grand Central Terminal (Fig. 3), which was finished early in



Figure 3. Grand Central Terminal, New York (Photo courtesy of Evelyn Weingardt)

1913. Wrote architectural chroniclers Pennoyer and Walker (2006), "The construction of Grand Central Terminal was one of the century's first important undertakings and most publicized building projects of the period, and it vaulted its architects, engineers and constructors into the national spotlight."

During his career, Balcom was responsible for more than a billion dollars worth of construction (in early 20th-century dollars), an astonishing accomplishment considering that a significant portion of his work was completed during the Great Depression. His contemporaries nicknamed him "the billion-dollar man." In today's dollars, the construction value of his projects would be in the \$22 billion range.

In the final analysis, Balcom was the classic "everyman," an ordinary soul who through his own efforts, diligence, and daring, along with random exposure to several remarkable circumstances and happenings, accomplished the extraordinary.

Homer was born on February 16, 1870, in Chili (Monroe County), New York, the only child of Mahlon and Francis (Gage) Balcom. Raised amid modest surroundings on the family farm near rural Kendall (Orleans County), New York, just a couple of miles from the southern shore of Lake Ontario, young Balcom was exposed to the virtues of honesty and hard work early in life. These lifelong qualities, as well as the inherited ambitions and resolve of generations of enterprising Balcoms, had a major influence on Homer as he matured into a confident adult—a determined but unassuming self-made man not afraid to roll up his sleeves and get the job done properly and on time. His credo seemed to be, Always do more than is expected of you, do it right, and never complain.

Homer's progenitors came from England. The first Balcom family member born in America (in Connecticut in 1697) was John Balcom, a sea captain first and a farmer second. John's children subsequently took up farming full time and became quite successful at it. Homer's great-grandfather, Asahel Balcom, along with his adult children, migrated from Connecticut to the Kendall area in 1816, the first Balcoms to settle in upstate New York. Farming remained the main occupation of most of them for several generations, although a few of the homesteading Balcoms also took up teaching or became affluent businessmen.

Homer's father Mahlon had abandoned the farm life in his early 20s to become a schoolteacher and then a retail dealer, which was his occupation when Homer was born. He was a middle-aged man of 39 (3 years older than his wife,

Francis) when their first and only child arrived. When he was just 1, Homer's grandfather (Mahlon's father) Daniel Balcom, a successful farmer and respected citizen of the area, suddenly died at age 69. It prompted Mahlon to return home to take care of his mother (Homer's grandmother) and run the family farm with his younger and only brother Walter, who was just in his teens.

Even though he was an only child and lived on a farm, young Homer was constantly surrounded by numerous cousins, aunts and uncles, and other close descendents of Asahel while he was growing up, all able and willing to offer advice and guidance. Never fond of the name "Homer" in his youth—and even later, when he was in college—Balcom went by "Gage," his middle name and the maiden name of his mother, whose father Moses Gage was a very successful doctor in the area. Later in adulthood, Balcom dropped "Gage" and went by "H. G."

Homer's parents and grandparents were confirmed advocates of the importance of formal education, and he was encouraged to get as much of it as possible. After finishing the highest level of schooling local Kendall schools had to offer, Homer enrolled at Brockport Normal School 10 miles southeast of Kendall, a short horseback or carriage ride from his home. At the time, Brockport Normal was well respected for turning out first-rate teachers and offering solid college preparatory courses. Incorporated in 1866, it officially became a college in 1942 and today is an integral part of the State University of New York (SUNY) system and one of SUNY's more selective colleges.

After taking Brockport's very difficult "classical program," re-

quiring 4 years to complete, Homer graduated with high marks in the spring of 1891. He was 21 years old; tall, lanky, and raw-boned; and eager to face the world and take on meaningful challenges. He immediately applied to a number of local schools for a position as a teacher, although his dreams for much more were fast developing. Those plans, though, had to wait. During the summer of 1891, while biding time until his first teaching assignment materialized, Homer helped his ailing father on the farm.

While Homer was still in his teens, a number of compelling events captured the nation's attention, as well as that of the citizens of Kendall, especially people interested in technological advances and construction. Among these events was the erection of increasingly taller monuments such as the 555.5-foot-tall Washington Monument obelisk, completed in 1884, the tallest masonry structure on earth, and the 984-foot-tall Eiffel Tower, finished 5 years later, the world's tallest metal structure.

Similarly, every time a new tall building or tower debuted in a major city like Boston, Chicago, New York, or Philadelphia (especially if bigger than some previous record holder), it was front-page news, hailed by the media as being something very special. The 1890 unveiling of New York's Madison Square Garden—with its 304-foot-tall observation tower, the "highest sightseeing point in Manhattan"—was one such news-capturing engineering feat:

Newspapers all over the country carried reports of the new building [Madison Square Garden]. It covered the entire block between Fourth and Madison avenues and 26th and 27th streets, encompassing an

area of 93,000 square feet, making it the largest amusement place in the country. As the press pointed out, the arena could accommodate more patrons than London's Albert Hall, the Paris Opera House, the Chicago Auditorium Building, and the Salt Lake City Mormon Temple. (Landau and Condit 1996)

Not to be overshadowed by all the late 19th-century high-rises capturing public attention—even in remote little towns like Kendall—were the nation's burgeoning number of newsworthy bridges, many of them international sensations and titleholders. Among the most iconic of these that would have been widely known during Balcom's youth were the 1883 Brooklyn Bridge over the East River (between Brooklyn and New York City), the 1889 Poughkeepsie Railroad Bridge over the Hudson River (near Poughkeepsie, New York), and the 1874 Eads Bridge over the Mississippi River (at St. Louis, Missouri).

The appearance of these well-engineered construction marvels suggested that many more engineering achievements were on the horizon and possible for anyone willing to take on the challenge of planning and building them. Becoming a civil/structural engineer to design such edifices certainly held much appeal for young people with high intelligence and ambition. Increasing numbers of bright and restless farm boys like Homer Balcom were showing great interest in the profession around the end of one of America's greatest centuries of industrial growth.

After much thought, in September 1892, an intense 22-year-old Balcom finally made a defining

life-changing move. Following his boyhood dream, he decided to become a builder of great engineering works. He quit his job teaching, bid his local friends and relatives adieu, and enrolled at Cornell University to study civil/structural engineering. Located in Ithaca, New York, 100 miles (as the crow flies) southeast of the Balcom farm in Kendall, Cornell was nestled in the picturesque hills at the southern tip of one of Central New York's narrow finger lakes, the long, 400-foot-deep Lake Cayuga.

On the first day of November 1893, 2½ months into his second year at the university, Homer's 62-year-old father Mahlon died. Homer rushed home, taking an indefinite sabbatical from his engineering studies, to put his father's affairs in order and make arrangements for his mother's future. He temporarily took a teaching position in the Kendall area while working out plans for returning to the university once he had set up his beloved and now completely alone mother on a place near his uncle and aunt (Walter and Melissa Balcom), who were by then in charge of the original Balcom family farm.

When Homer re-entered Cornell in the fall of 1894, he was a year older and considerably more mature, certainly, than most of his classmates. Even though he made regular trips back home to check on his mother, first and foremost on his mind and agenda was mastering his engineering courses. In addition to his studies, he was involved with a number of engineering and student groups, often in leadership roles. The caption under his senior year photograph in the 1897 Cornell yearbook (*The Cornellian*) read, "Throughout his four years, Homer Gage Balcom

has been recognized as one of the best men in his department. In his senior year he was honored with an election to Sigma Xi [the honorary scientific research society]."

During his final semesters on campus, Balcom encountered two fascinating undergraduates, both a few years behind him in school, who would one day play a major part in his career—Wilton Joseph Darrow (1876–1945), an outgoing civil engineering student who participated on Cornell's lacrosse and football teams, and Richmond Harold Shreve (1877–1946), a promising architectural student who became professor of architecture at the university and then founded a highly successful architectural firm in New York City.

After earning his degree in civil engineering in 1897, at age 27, Balcom left upstate New York for good, moving to East Berlin, Connecticut, where he went to work for the Berlin Iron Bridge Company as a draftsman and junior staff engineer. At the time, Berlin Iron was one of the largest employers in the state, world famous for its unique, patented bowstring-shaped bridges and its skill at fabricating and building various types of complicated steel and iron bridges and roof structures.

Shortly after arriving in East Berlin, Balcom took up residency at a rooming house owned by Joseph McCracken, an ironworker associated with Berlin Iron. Among the other four boarders, all single, were a 26-year-old architect, a bridge builder, a bridge construction foreman, and a bridge engineering superintendent—all three of the latter in their early 40s. The five had much in common, but of much more personal interest to Balcom was McCracken's attractive, unmarried stepdaughter, Gertrude McCrum.

She was the daughter of McCracken's wife of 3 years, Jennie, who was the widow of a successful East Berlin manufacturer, William McCrum.

The earnest, bespectacled, and up-and-coming structural engineer and the landlord's charming stepdaughter quickly hit it off and increasingly spent more of their free time together. In due course, their relationship developed into a romance, and they began talking about the future.

Always an eager and quick learner, Balcom's work assignments at Berlin Iron rapidly increased in technical difficulty, accountability, and importance. In early 1900, Berlin Iron, along with numerous other bridge companies around the country, was suddenly purchased, taken over, and absorbed by the American Bridge Company (ABC), a conglomerate of J. P. Morgan. Balcom was the right man in the right spot at the right time. He was quickly promoted to design engineer with a hefty pay increase.

Before the year was out, Balcom's relatively tranquil existence mostly doing standard projects and experiencing small-town life came to an abrupt end as he was thrown into the fast-moving world of industrial turn-of-the-century America. Driving much of the progress was the explosive use of structural steel for complex bridges and buildings being pushed by U.S. industrialists like Morgan and Andrew Carnegie.

Along with the drastic change in Balcom's work assignments came an end to his bachelorhood as well. On October 24, 1900, in the midst of brilliant foliage on a multicolored Connecticut fall day, he and Gertrude married. Both were 30 years old. By then, they were only vaguely aware of

where his work at ABC would take him and how much away-from-home travel would be involved. They would learn soon enough.

In May 1901, Balcom was assigned to New York City. Then, in December 1901, he was again promoted and relocated to Philadelphia. From 1902 until 1904, he was “engineer in charge of design” in ABC’s New York City office. Later in 1904, he was made engineer in charge of all of the company’s design work and transferred to Pittsburgh, nearer many of the company’s key operations. Working directly under Balcom at the time was a former engineering school underclassman from his Cornell days, Wilton Darrow, who for most of his time at ABC served as Balcom’s assistant engineer.

At the same time Balcom was going through all his promotions and relocations at ABC, major plans were underway in New York City for building a gigantic new state-of-the-art railroad interchange in the middle of Manhattan Island—Grand Central Terminal. It was the world’s most elaborate and sophisticated transportation complex ever conceived. Selected as architect for the project was Reed and Stem, a firm from St. Paul, Minnesota, well versed in the design of leading-edge railway terminals. In the fall of 1903, shortly after Reed and Stem was chosen (after an exhaustive design competition), the politically and socially well-connected New York architect Whitney Warren, head of Warren and Wetmore, approached William Vanderbilt, CEO of New York Central Railroad (owner of the project), to get his firm added to the Stem and Reed team as copartners. Warren and Vanderbilt were close acquaintances as well as relatives of sorts, and

Vanderbilt had few reservations about adding Warren’s firm to the Grand Central Terminal architectural team, to the dismay of Reed and Stem. In February 1904, Reed and Stem and Warren and Wetmore became the Associated Architects of Grand Central Terminal. Charles Reed was designated as chief executive of the unholy alliance.

Once Associated Architects had established its offices at 314 Madison Avenue, where both architectural firms kept separate books, Reed began searching for qualified engineers for the team. By June 1905, Balcom had left ABC and was on board, in charge of the alliance’s structural engineering department. To aid him, he brought along his assistant engineer from ABC, Wilton Darrow.

The mastermind behind the planning of Grand Central Terminal was New York Central’s chief engineer and vice president in charge of construction, William Wilgus (1865–1949), one of America’s most renowned railroad engineers. His visionary terminal plan called for an elaborate double-level rail station (commuter trains on one level and long distance trains on the other) buried two stories below ground with the terminal facility directly above—along with several high-rise buildings, which were built using the concept of “air rights” leasing, a Wilgus invention. He referred to this innovative concept as “taking wealth from the air.” Wilgus’s scheme not only opened up several blocks of Manhattan real estate for profitable development, it also restored dozens of crosstown streets previously divided by surface rail tracks and removed from public view many of the area’s unsightly railyards. His leasing

concept for over-the-rail-tracks buildings significantly helped offset and finance the immense construction cost of the terminal center: “The terminal building alone cost \$43 million in 1913, the equivalent of nearly \$750 million [\$1 billion in 2011 dollars] today” (Schlichting 2001).

Constructing Grand Central, with its complicated underground railyard, numerous interrelated utilitarian structures, and tall buildings overhead, required an intricate network and complex maze of structural systems. In designing them, Balcom came up with several first-time-ever-used framing systems that not only spanned railroad lines but also dealt with track vibrations and handled unstable foundation conditions. To insulate above-the-ground building frameworks from vibrations and stresses caused by track and street movements, he made use of a number of different types of independent foundation systems, large vibration-absorbing footing mats, and unique adjustable separators between buildings and traffic-carrying structures.

The first multistory building engineered by Balcom and Darrow and constructed within the 12-block “terminal city” area using New York Central’s “great air rights plan” was Grand Central Post Office in 1909. Among others that came later was the 24-story Biltmore Hotel in 1912, Balcom’s tallest building at the time.

When Balcom’s daughter Gertrude Frances Marie was born on January 30, 1907, he was living in an upscale townhouse at 3 Hamilton Avenue in Yonkers, New York. Both Balcom and his wife were 37 at the time, about the same age his parents were when they had him. Gertrude Francis would be the couple’s only child.

Less than a year later, in January 1908, Balcom split from Associated Architects and formed a private-practice partnership with his longtime friend and colleague Wilton Darrow. Simply branded Balcom and Darrow, the newly established consulting structural engineering firm listed Balcom as senior partner and the 6-year-younger Darrow as junior partner. Not married at the time, Darrow would remain a confirmed bachelor his entire life.

Even though Balcom and Darrow were no longer employees of Associated Architects, their new consulting offices were located right next door to the alliance, at 314 Madison Avenue, and they continued to do structural engineering on the Grand Central project. Balcom's firm also quickly built strong relationships with many of New York's leading architects, including Warren and Wetmore, Reed and Stem, Schultze and Weaver, York and Sawyer, Voorhees-Gmelin and Walker, Raymond Hood, and Carrere and Hastings, where one of his old Cornell acquaintances, Richmond Shreve, was currently the senior partner. In 1922, Shreve would establish his own architectural practice with another principal from Carrere and Hastings, William Lamb. Later, they would add a third partner and rename their firm Shreve, Lamb and Harmon.

Shortly after opening his office in 1908, Balcom, along with Shreve and several dozen other upward-moving and successful New York executive types, purchased property and built large, comfortable manor homes in the exclusive Riverview Manor subdivision in the rolling hills of Hastings-on-Hudson in Westchester County, a short trainride commute from downtown Manhattan. Only a

few lots apart, the impressive homes of Balcom and Shreve were high up, overlooking a picturesque stretch of the Hudson River, which had on its far bank majestic granite palisades. Every morning when they rose and in the evenings when home from work, Balcom and Shreve could enjoy both the idyllic wooded countryside and unobstructed views of a rapidly growing New York City skyline, where many of their buildings were poking higher and higher into the sky. Just two decades later, working together, the pair would build the tallest and most dazzling structure in history.

Once he was well situated in Hastings-on-Hudson, which in many ways reminded him of Kendall and his youth, Balcom became involved with numerous community activities. He was, for example, a leader and firefighter in Local Unit No. 3 (Fig. 4) and, in January 1912, was elected to the Hastings Board of Education. From 1921 to 1933, he held the office of president and, as such, exercised great influence in the development of a modern suburban school system. He created a centralized school system where children from all sections of the village, both industrial and residential, received a solid education.



Figure 4. Balcom (tall man standing behind fire truck in center of photograph) in his firefighter gear (Photo courtesy of Lois Moon Welch)

When Grand Central Terminal officially opened to the public at midnight on February 1, 1913, amid great celebration and scores of officious speeches, many were present to take bows. Who immediately stepped to the forefront, taking virtually all honors for the building's design, was the architectural firm of Warren and Wetmore: "Despite the fact that Reed and Stem had initially won the commission and was vital to the project's planning and success, Warren and Wetmore was more assertive in taking credit for its design and claiming the honors" (Pennoyer and Walker 2006). Few attending the ceremonies heard the names of any of the engineers involved, including Wilgus, Balcom, or Darrow, who were so crucial to the project's monumental technical accomplishments. Even today, they remain unknown to most.

Right after the grand opening of Grand Central came the debut of the first of many significant Balcom-designed high-rises located outside of "Terminal City"—the 20-story Gramercy Park Office Building (Fig. 5). It marked the beginning of what would be a very long line of noteworthy skyscrapers Balcom would design. At the same time, though, his partnership with Darrow was showing signs of malfunction. Always independent and flighty, the firm's bachelor partner was restless and spending more and more office-hour time on other pursuits, one of them being a religious movement, the Rosicrucian Fellowship. Right around the time Gramercy opened, Darrow took time off from the firm and sailed to England. Upon his return, he became even more engrossed in Rosicrucian activities.

In 1916, Darrow officially "retired" from Balcom and Darrow,



Figure 5. Gramercy Park Building, New York (Photo courtesy of Evelyn Weingardt)

and the successful 8-year-old partnership was dissolved. Darrow took a position as structural engineer with the American Can Company a few blocks away from 314 Madison Avenue, while Balcom continued on as a consulting structural engineer, renaming his sole-proprietor operation H. G. Balcom. By then, Balcom had a solid international reputation for designing impressive multi-story structures, mostly using structural steel.

Less than a year later, after the United States declared war on Germany on April 6, 1917, and entered World War I (1914–1918), Balcom stepped forward to help the war effort. In addition to keeping his consulting engineering practice operational, Balcom signed on as chief structural engineer for the American International Shipbuilding Corporation (AISC), fabricating warships for the U.S. government at the Hog Island Shipyards in Pennsylvania. AISC was a wholly owned subsidiary of the American International Company, whose board of directors was a virtual Who's

Who of American industry and finance. The initial stages of AISC's emergency fleet-building operations were headed by none other than America's most famous civil engineer of the time, Gen. George Goethals (1858–1928), fresh from his successful completion of the Panama Canal.

As the largest shipyard in the world at the time, Hog Island built 122 military vessels for the war effort, the most by a wide margin of any shipyard. In addition to Balcom, numerous other Cornell graduates were working in key positions at Hog Island during the war, so many that a special AISC Cornell Club was formed. As the war began winding down, members of the club presented Balcom with a formal certificate of appreciation for his stellar leadership while chiding him for his chain-smoking habit, which never interfered with his work.

When World War I ended, Balcom returned full time to running his engineering company, which was overflowing with lucrative postwar projects being fueled by the seemingly endless exuberance and optimism of the Roaring Twenties. Balcom's projects and his client list grew right along with the nation's booming economy; each succeeding high-rise structure he designed seemed bigger, taller, or more important than the previous one.

Those glory days lasted for over a decade until "Black Thursday" and the stock market crash of 1929. Balcom's last major skyscraper completed just before the crash was the 35-story-tall 230 Park Office Building at Grand Central Terminal, the headquarters for New York Central Railroad.

Shortly after Wall Street's financial collapse and the start of the

Great Depression, Balcom was involved in a fantastic and dramatic three-way race to build the world's tallest building, an American structure even taller than the Eiffel Tower in France. Front and center, and deeply committed to that race, were three of New York's leading architectural firms and three of its most daring structural engineering firms, all ready and willing to put their reputations on the line. All were sure they could break the elusive 1,000-foot mark.

Lead architect for the project that was completed first, the Bank of Manhattan Trust Building (a.k.a. 40 Wall Street Building), was H. Craig Severance. His structural engineer was Purdy and Henderson, legendary skyscraper engineers who, over its history dating back to its founding in Chicago in 1885, had designed more tall buildings than any engineering company of the era. From 1909 until 1913, when the 792-foot-tall Woolworth building was completed and crowned the world's tallest, Purdy and Henderson's 700-foot-tall Metropolitan Life Tower held the crown. Purdy and Henderson's founder Corydon Purdy (1859–1944) was eager to once again be recognized as the structural engineer for the tallest skyscraper in existence. The second group of competitors, the Chrysler Building team, included William Van Allen as architect and the father-son team of Ralph and William Squire as structural engineers. The owner-developer for the project was the cagey automobile tycoon Walter P. Chrysler.

Severance and Van Allen, both with massive egos, had once been partners but were now mortal enemies, detesting the mere mention or sight of one another. And Chrysler himself was without peer when it came to pulling out all

stops in besting any and all competitors, and he was pushing Van Allen and the Squires at every turn to win the tallest-building race, whatever the cost.

The third building in the race, slightly behind the other two in schedule, was the Empire State Building, a purely speculative venture put together by two of the nation's most prominent entrepreneurial industrialists, John Jacob Raskob and Pierre du Pont. Former New York governor and U.S. presidential candidate Alfred E. Smith was also involved and was the project's front man, its public face. Balcom's Hastings-on-Hudson neighbor Richmond Shreve of Shreve, Lamb and Harmon led the building's architectural team. The general contractor was Starrett Brothers and Eken Inc., and the structural engineer was H. G. Balcom.

There was no question that the strong-minded Raskob, rather than du Pont, was the group's main leader. His was a true rags-to-riches success story, coming from poverty and ending up as a powerful vice president at General Motors and making a fortune along the way. He, too, had an oversized ego, and he demanded that his design-construction team give him the greatest skyscraper ever built, "the tallest structure that could be made to stand."

The intense three-way race unfolded this way: The 40 Wall Street team was privy to Chrysler's construction drawings showing its final projected height. They were not, however, privy to secret plans to erect a large 185-foot spire above the building height shown on the drawings, something the Chrysler team schemed to do during the darkest of night.

The 40 Wall Street project finished construction first, early in

1930, and recorded its record-making building height as 927 feet (from street to the tip of its pinnacle), several feet taller than the height they were sure Chrysler's building was going to be. In publicly announcing they now held the world record, there was much fanfare and great celebration among the executives of the bank, overjoyed that they had finally outdone the rascally Walter Chrysler. Severance himself was wildly ecstatic because he had bettered his despised ex-partner Van Allen. And Purdy was popping his buttons knowing he once again held the tallest building structure title. But their jubilation was short lived.

A month after 40 Wall Street opened, the Chrysler team unveiled its enormous hidden spire, dramatically extruding it up through a roof opening, then quickly attaching it to the building top. It established the Chrysler Building's height at 1,050 feet, the first structure in history to rise above 1,000 feet. But nearby, the third participant in the three-way contest was still growing *and growing*. Rumors were fast spreading that it would be the first building to have more than 100 stories!

Then, on the first day of May 1931, all the year-long bickering that had been going on between Severance and Van Allen, and between Chrysler and Bank of Manhattan officials, about having the tallest edifice was put to rest. The world now definitely had its new world record holder—the Empire State Building. Standing 102 stories and 1,250 feet tall, and towering over everything around it, the structure was officially opened for business by President Herbert Hoover when, from the White House, he pressed a button that turned on the building's lights, amid great

cheering and celebration. Budgeted at \$43 million, the project cost only \$24.7 million because it was built during the Depression. And for the same reason, it immediately fell on hard times and didn't reach 85% occupancy until 1944.

Three other New York City signature high-rises engineered by Balcom's firm came on the scene during the Depression, including the 50-story Bank of New York Building, the 47-story Waldorf-Astoria Hotel (Fig. 6), and the 69-story GE Building at Rockefeller Center Plaza (Fig. 7). By the early 1930s, Balcom's firm had grown so much he needed more executive power and administrative help at the top. He was never going to have another name partner, not after his Darrow experience, but he was OK with junior partners or associates. To accommodate them, he renamed his company H. G. Balcom and Associates.

Although many of Balcom's projects were in the state of New York, his practice spanned the globe. Among his significant



Figure 6. Waldorf-Astoria Hotel, New York (Photo courtesy of Sal Galletta)



Figure 7. GE Building (background) at Rockefeller Center Plaza, New York (Photo courtesy of Sal Galletta)

international projects were the Louvain University Library in Belgium, the Devonshire House in London, and the Young Men's Christian Association (YMCA) building in Jerusalem. His U.S. projects beyond New York City included the National Gallery of Art (Mellon Art Museum), U.S. Archives, and Department of Commerce buildings in Washington, D.C.; the distinctive U-shaped Ingraham Building in downtown Miami, Florida; the Cathedral of Learning at the University of Pittsburgh; the First National Bank of Boston; and the Nebraska State Capitol in Lincoln (Fig. 8). Especially unique about the Nebraska structure—in addition to its 400-foot height, making it the second tallest state capitol in the country behind Louisiana—was its tower, a narrow golden dome with an intriguing statue on top. Created by artist Lee Lawrie, the 19-foot-tall bronze sculpture *The Sower* depicts a farmer scattering seed, representing the state's rich agricultural heritage. Because of its tall, narrow silhouette, the Nebraska Capitol building is often referred to as



Figure 8. Nebraska State Capitol, Lincoln, Nebraska (Photo courtesy of David R. Weingardt)

“The Tower of the Plains.”

Representative of Balcom's bridge projects was the Sixteenth Street Bridge (Fig. 9) over the Allegheny River in Allegheny County, Pennsylvania, which opened on October 9, 1923, at a cost of \$1.25 million. The 2,000-foot-long, four-lane bridge, with pedestrian walkways on both sides, is a steel-trussed arch structure with three main spans over the river. The center arch spans 437 feet, while the north and south arches span 240 feet. The bridge's approach spans are riveted plate girders and bents. The previous



Figure 9. Sixteenth Street Bridge over the Allegheny River, Pittsburgh, Pennsylvania (Photo courtesy of Richard Weingardt Consultants)

bridge at the location was the last standing wooden covered bridge in the city, which was destroyed by fire in 1919.

Throughout his career, Balcom consistently gave back to his profession and community. In addition to his Hastings-on-Hudson activities, he was a past master of the Fernbrook Lodge of Masons and a great supporter of both of his alma maters, Brockport College and Cornell University. Not only a frequent contributor and visitor to Cornell, he also often lectured and presented papers there. He was a leader in the Cornell Society of Engineers and Sigma Xi, and he served as a director of the New York State Society of Professional Engineers and was president of its New York City Chapter. He was a longtime member of and leader in the American Society for Testing Materials and the American Institute of Steel Construction, for which he chaired the 1928 development of new standards for fireproofing steel frames. For ASCE, he chaired the committee producing its special 1929 Hurricane Report. An expert on several subjects, Balcom was always willing to assist fellow engineers with advice whenever approached, and more than any engineer of his time, he helped promote public recognition of the engineering profession.

Balcom passed away peacefully on Sunday night, July 3, 1938, in Hastings-on-Hudson, at age 68. He had checked into the local hospital 4 days earlier for a heart ailment. Masonic funeral services were held at his residence at 65 Calumet Avenue, and immediately after, he was buried at Sleepy Hollow Cemetery, Tarrytown, New York. He was survived by his wife Gertrude, his daughter Gertrude Francis Marie Moon



Figure 10. Addition of television tower to the top of the building's spire in 1952, which increased the Empire State Building's overall height from 1,250 to 1,475 feet (Photo courtesy of Lois Moon Welch)

and her husband John Moon, and one granddaughter, Barbara Joan Moon. (Twins Lois Gage Moon and John Andrew Moon, Jr., arrived 7 months after their grandfather's interment.)

Fourteen years after his passing, Balcom's ultimate masterpiece grew once again in height, from 1,250 to 1,475 feet, with the addition of a television tower to the top of the building's spire (Fig. 10).

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